Essays on Applied Econometrics

Dissertation Defense

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Overview

1. **Targeted Credit Access and Entrepreneurship**
   Can targeted financial access, via specialized lender programs, improve entrepreneurial activity in underserved communities?
2. ***Ce modèle vous plaît?***
   How can experimental preference data on LLMs be leveraged to design cost-efficient, heterogeneity-aware routing policies?
3. **Single-Shot Effects and Semiparametric DiD**
   How can we identify and estimate impulse-response parameters in DiD settings with time-varying treatments and covariates?

Outline of Talk

1. Chapter 1: Targeted Credit Access and Entrepreneurship
2. Chapter 2: *Ce modèle vous plaît?*
3. Chapter 3: Single-Shot Effects and Semiparametric DiD
4. Conclusion

Entrepreneurship Gaps in the United States

- **Persistent gaps in entrepreneurship:** Business formation remains substantially lower among minority and low income groups and neighborhoods (Fairlie 2022)
- **Capital access frictions as a central mechanism:**
  - **Discrimination and banking access** Differential loan approval rates and credit rationing (Blanchflower et al. 2003; Chatterji and Seamans 2012; Li 2022)
  - **Credit histories and expectations** Weak credit worthiness, collateral, and access perceptions (Fairlie 2022; Bennett 2024)
- **Why it matters:** Financial frictions limit entry of high ability potential entrepreneurs, reduce firm scale and survival, and constrain inclusive economic growth

Motivated Policies

- Policymakers aim to ease capital access constraints to unlock local employment and neighborhood development by enabling entry and growth of constrained entrepreneurs.
- **Common tools:**
  - **Disclosure regulations** - e.g., Community Reinvestment Act
  - **Credit guarantees and subsidies** - e.g., Small Business Administration loan subsidies (504 and 7a programs)
- **Limitations:** Disclosures influence behavior only indirectly, guarantees and tax incentives may not be tightly targeted to the most constrained or highest return entrepreneurs
  - Hackney et al. 2023; Ding and Lee 2018

Targeted Place-Based Lending via CDFIs

- **A targeted, place-based approach:** Expand credit access via mission-driven lenders that specialize in constrained borrowers and disadvantaged neighborhoods.
- **Community Development Financial Institutions (CDFIs):**
  - Increases credit supply by certifying and subsidizing community-oriented lenders
  - Intermediaries include nonprofit banks, credit unions, and loan funds operating in designated markets
- **What CDFIs provide:**
  - **Targeted lending:** majority of financing directed to defined populations and places
  - **Diverse intermediary services:** consumer, housing, small business, and venture finance; advisory and technical assistance

This Paper

- **Objective:** Evaluate how local access to CDFIs affects the rate, composition, and early outcomes of new startups
- **Motivation:** Nearly three decades in, causal evidence on CDFI program effects remains scarce (Harger et al. 2015; Johnson et al. 2025; Gong et al. 2023; Swack et al. 2015)
- **Empirical Strategy:** Employ a difference-in-differences design using ZIP-level entrepreneurship outcomes, with treatment defined as the observed entry of CDFIs from disclosure reports

Background

CDFI Program

- Established under the 1994 Riegle Community Development and Regulatory Improvement Act and overseen by the U.S. Treasury
  - First certifications and awards in 1996 with 69 lenders
- Participating lenders receive competitive grants from the CDFI Fund to support lending and technical assistance
- Certified loan funds often leverage CRA motivated partnerships with banks for supplemental capital and investment
- **Program size today:**
  - More than 1,400 certified CDFIs nationwide across all 50 states
  - Estimated $500 billion in managed loan portfolios, as of 2023 (Anderson 2024)

Target Investment Markets

- To maintain certification, lenders must direct at least 60% of financing activity toward a designated investment market

- At certification, CDFIs designate one or more target markets and must annually demonstrate sufficient activity within them

- Target markets can be:
  - **Geographic (Investment Areas):** Census tracts that meet at least one criterion poverty rate $\geq 20\%$, median income $\leq 80\%$ of the U.S. median, or unemployment $\geq 1.5\times$ the U.S. rate
  - **Population based (Targeted Populations):** Defined borrower groups and purposes, such as Native communities, minority borrowers, or other underserved populations

Data

Data Sources

Balanced panel of $\approx 32{,}000$ ZIP codes from 1988 to 2016

1. **CDFI Fund Transaction Level Reports (TLR)**
   - Loan-level data on CDFI activity after CDFI Fund grant receipt
2. **Startup Cartography Project**
   - ZIP code measures of entrepreneurial activity based on state business filings and prediction models
   - Includes corporations and partnerships, excluding non-profits, non-employer, and sole-proprietor startups
3. **Demographic and other control variables**
   - Decennial Census, National Neighborhood Data Archive, ZIP Code Business Patterns

Outcomes of Interest

- **Startup formation**
  - (Flow) Startup formation rate: new firms per 1,000 residents
  - (Stock) Number of Establishments per 1,000 residents
- **Startup quality**
  - Average predicted startup quality score from the Startup Cartography Project models
- **Growth outcomes**
  - Counts of high growth events such as IPOs and acquisitions within 6 years of founding among local startups
- **Labor market outcomes**
  - Employment-to-Population ratios and log wages

Empirical Strategy

Local CDFI Access and Startup Activity

- Study how local startup formation responds when neighborhoods gain access to at least one CDFI lender
- Use a difference in differences design with staggered timing of CDFI entry across counties
  - Infer CDFI entry from the first appearance of TLR lending
  - Assign treatment cohorts at the county level
  - Study outcomes at ZIP level, study demographic heterogeneity

Ideal Reduced Form Model

Two-Way Fixed Effects:

$$Y_{ict} = \alpha_{ic} + \delta_t + \beta \mathbb{I}[\text{ZIP } i \text{ in a CDFI's active TIM}] + \varepsilon_{ict}$$

- Y_{ict}: entrepreneurship outcome for ZIP i in county c at time t
- Treatment indicator is whether ZIP i is (at least partly) included in an existing CDFI's active investment market
- **Main Identification Challenges:**
    1. Do not observe individual CDFI's TIM definitions nor their exact certification/entry date
    2. CDFI entry is non-random and may be related to other factors that afffect entrepreneurial activity

Feasible Reduced Form Model

Alternative Specification:

$$Y_{ict} = \alpha_{ic} + \delta_t + \tilde{\beta}\mathbb{I}[\text{Observed Lending in County } c \text{ at some } t' \leq t] + \tilde{\varepsilon}_{ict}$$

- Define treatment as the observed ramp-up of CDFI activity in a local credit market following subsidy receipt from the CDFI Fund
  - Local credit market proxied for by county-level definition.
    $\rightarrow$ Finer definitions may mis-classify treatment
- TLR disclosures reveal all award-associated lending
  - **Operating Assumption:** first recorded areas of lending reveal lenders' approximate investment market/geographic scope
  - Over 90% of CDFIs receive one CDFI-Fund-backed award

Addressing Non-Random Entry

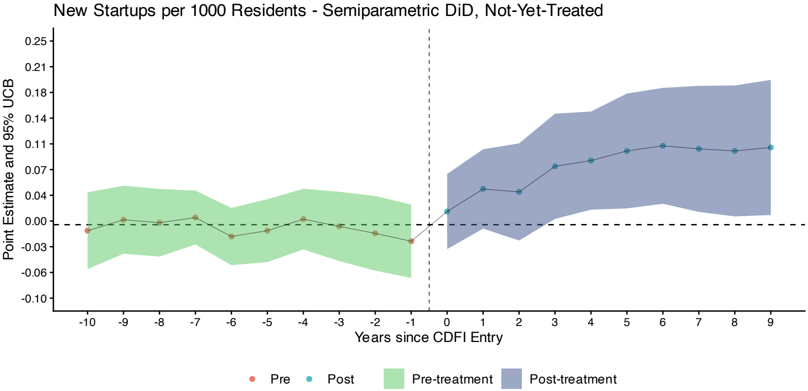
- Identification of $\tilde{\beta}$ may still be affected by other local factors impacting both entrepreneurship trends and CDFI entry
  - ZIPs in counties where CDFIs never enter are likely poor comparisons due to demographic differences
  - Instead, eventually-treated ZIPs come from the same policy-eligible pool, with similar institutional motivations for CDFIs to define them as a TIM
  - Because we define treatment at a coarse level (county) rather than unobserved target markets for each lender, it is important to compare neighborhoods with similar characteristics

Estimation

- Estimate each $ATT(g,t)$ and aggregate to event-study effects, following Callaway and Sant'Anna (2021)
- Employ doubly-robust $ATT(g,t)$ estimators
  - Large covariate set $\rightarrow$ apply the Double Machine Learning (Chernozhukov et al. 2018) to estimate first-stage outcome regression and propensity score nuisance parameters
  - Use cross-fit random forests – better validation set performance than LASSO or LightGBM
  - Cluster at state level, report uniform confidence bands

Results

CDFI Entry and Startup Formation



Notes: Standard errors clustered by state. First stage estimates obtain using cross-fit random forests.

Tabulated Results

Outcome	10-Year ATT (SE)
New Startups per 1,000 Residents	0.11* (0.03)
Establishments per 1,000 Residents	0.17* (0.04)
Avg. Predicted Quality of New Startups	0.00 (0.02)
Number of Growth Events (IPO/Acquisition)	0.00 (0.01)
Employment-to-Population Ratio $(\times 100)$	0.47* (0.11)
Log Average Wages	0.00 (0.00)

Notes: * $p < 0.05$. Standard errors clustered by state.

Demographic Heterogeneity in Startup Formation

Subgroup	10-yr ATT (SE)
High Minority (bottom quartile % White)	0.12* (0.04)
Low Minority (top quartile % White)	0.06 (0.05)

Notes: Outcome is new startups per 1,000 residents. * $p < 0.05$. Standard errors clustered by state.

Discussion

Discussion

- **Targeted credit works on the extensive margin.** CDFI entry brings in more founders without lowering predicted quality or realized growth outcomes.
- **The new activity is real and durable.** Establishment counts and local employment rise and persist, indicating net new business formation rather than reshuffling across places or firms.
- **Effects are largest where credit constraints bind.** Gains concentrate in high-minority, lower-income areas, consistent with the program's targeting and with models of credit-constrained entry (Evans and Jovanovic 1989).

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Large Language Models and Task Deployment

- **Context.** Organizations deploy many LLMs across heterogeneous tasks. Models differ greatly in task-specific quality and inference costs.
- **The Problem.** Which model should serve a given task?
- **Why It Matters.** Inference at scale is cost-intensive. Always routing to the best-benchmarked model can be inefficient.

This Paper

- **Objective:** Estimate *heterogeneous* LLM quality measures from experimental preference data and use it to design cost-aware routing policies
- **Approach:** A heterogeneity-enriched Bradley–Terry model whose parameters vary with prompt embeddings, estimated by structural deep learning (Farrell, Liang, and Misra 2021)
- **Routing Policy Design:** Embed model-specific quality and cost prediction functions into a planner's routing problem; learn the policy via Athey and Wager (2021), with an orthogonal score
- **Empirical Application:** 48,760 pairwise evaluations across 50 LLMs from the French government's Compar:IA platform, with per-query energy/token costs

Model Framework

Classical Bradley–Terry Model

In practice. Platforms like Chatbot Arena (Chiang et al. 2024) crowdsource *randomized pairwise* comparisons of LLM responses and rank models from the human votes via Bradley–Terry.

Standard formulation (Bradley and Terry 1952; Luce 1959). Each model m has a scalar quality parameter $\theta_m \in \mathbb{R}$. The win probability for model i over j is:

$$\Pr(Y_{rk}(i,j) = 1) = \frac{\exp(\theta_i)}{\exp(\theta_i) + \exp(\theta_j)}$$

Properties. Transitivity across alternatives. Only pairwise quality differences $\theta_i - \theta_j$ are identified $\Rightarrow$ normalize $\theta_1 = 0$.

Limitation. Preferences are **homogeneous**: relative quality does not vary across prompts, tasks, or contexts. A model that is best on average may be mediocre for specific prompts.

Heterogeneity-Enriched Bradley–Terry

Enrichment: replace the constant θ_m with a flexible function of the conversation embedding X_k (Li and Li 2025; Zhang et al. 2025).

$$\Pr\left(Y_{rk}(i,j) = 1 \mid X_k\right) = \frac{\exp\{\theta_i(X_k)\}}{\exp\{\theta_i(X_k)\} + \exp\{\theta_j(X_k)\}}$$

The functions $\{\theta_m(\cdot)\}_{m\in\mathscr{M}}$ are defined as loss minimizers

$$\{\theta_m(\cdot)\}_{m\in\mathscr{M}} = \arg\min_{\{f_m\}} \mathbb{E}\left[\ell\left(Y_{rk}, f_{I_{rk}}(X_k), f_{J_{rk}}(X_k)\right)\right]$$

where ℓ is the Bradley–Terry logistic log-likelihood.

Each embedding X_k induces *different* preference parameters and potentially *different* model rankings.

Identifiable Parameters

Average Preference Parameter:

$$\bar{\theta}_m = \mathbb{E}[\theta_m(X_k)]$$

Summarizes the average quality of model m relative to the baseline, integrating over prompt contexts.

(Average) Dominance Probability:

$$Q_m(X_k) = \frac{\exp\{\theta_m(X_k)\}}{\sum_{j=1}^{M} \exp\{\theta_j(X_k)\}}, \qquad \bar{Q}_m = \mathbb{E}[Q_m(X_k)]$$

i.e., the probability that m is most preferred among M alternatives

Estimation and Routing Policies

Estimation: Structural Deep Learning (Farrell, Liang, and Misra 2021)

- Estimate $\{\theta_m(\cdot)\}_{m=2}^{M}$ with a **multi-headed neural network**:
  - *Shared trunk*: maps $X_k \in \mathbb{R}^d$ through a shared network
  - *Model-specific heads*: each head m outputs $\hat{\theta}_m(X_k) \in \mathbb{R}$
- Construct a Neyman-orthogonal score $\psi(W_{rk})$ for a given smooth functional $\mu = \mathbb{E}[G(X_k, \theta(X_k))]$ of the preference parameters (and other regression objects):

$$\psi(W_{rk}) = G(X_k, \theta(X_k)) - G_\theta(X_k, \theta(X_k)) \underbrace{\mathbb{E}\left[\ell_{\theta\theta}(Y_{rk}, \theta(X_k)) \mid X_k\right]^{-1}}_{\text{inverse conditional Hessian}} \ell_\theta(Y_{rk}, \theta(X_k))$$

- Neyman orthogonality facilitates valid statistical inference and policy learning (Athey and Wager 2021)

A Cost-Aware Routing Problem

Setup. For each conversation k with embedding X_k, the planner chooses a model to deploy.

Inference costs: $c_m(X_k) = \mathbb{E}[K_m \mid X_k]$, where K_m is the energy/tokens/money consumed by model m

Routing policy: $\pi : \mathscr{X} \to \mathscr{M}$. Maps from prompt context to model.

Planner's objective: minimize expected inference cost conditional on selecting the user-preferred model (i.e., re-weighted cost minimization):

$$\boxed{\min_{\pi \in \Pi} \mathbb{E}\left[c_{\pi(X_k)}(X_k) \,\middle|\, M^* = \pi(X_k)\right] = \min_{\pi \in \Pi} \mathbb{E}\Big[\frac{Q_{\pi(X_k)}(X_k)}{\mathbb{E}\left[Q_{\pi(X_k)}(X_k)\right]} c_{\pi(X_k)}(X_k)\Big]}$$

Intuition. Route to cheaper models when dominance probability differences are small; use expensive models only when they are strongly preferred for a given prompt.

Policy Learning

Augmented loss. To construct orthogonal scores for the ratio objective, jointly estimate preferences *and* inference costs:

$$\ell^*(W_{rk}, \varphi) = \ell_{\mathsf{BT}}(W_{rk}, \theta) + \sum_{m=1}^{M} \tfrac{1}{2}\big(K_m - c_m(X_k)\big)^2$$

where $\varphi = (\theta, c)$ collects all nuisance functions and $c_m(X_k) = \mathbb{E}[K_m \mid X_k]$.

Orthogonal influence function. For each model m, construct an orthogonal score $\psi_m^*(W_{rk})$ for $\mathbb{E}[c_m(X_k) \mid M^* = m]$ using results from Farrell, Liang, and Misra (2021).

Policy optimization (Athey and Wager 2021):

$$\max_{\pi \in \Pi} \mathbb{E}\left[\sum_{m=1}^{M} -\psi_m^*(W_{rk})\, \mathbf{1}\{\pi(X_k) = m\}\right]$$

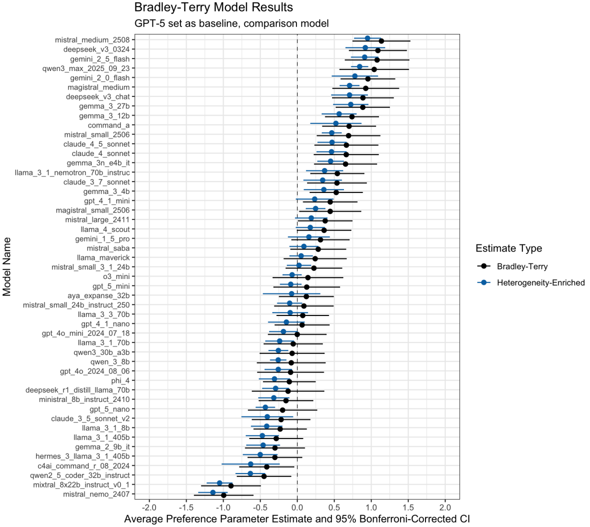
Policy class: neural network classifier with max-of-softmax output.

Empirical Application

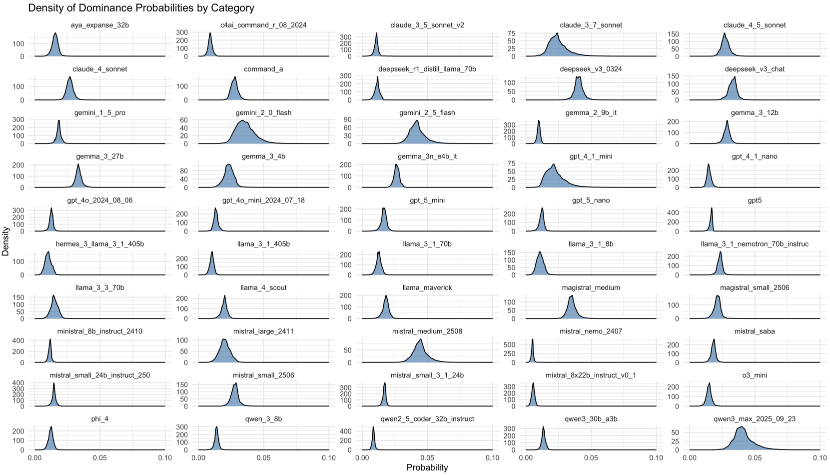
Data: The Compar:IA Platform

- **Setting.** The French government's Compar:IA platform: large-scale, randomized pairwise human evaluation of 50 LLMs, paired with model-level inference-energy costs.
- **Source.** Compar:IA, a large-scale pairwise-evaluation platform run by the French Ministère de la Culture: randomized assignment with blind outputs and post-vote reveal.
- **Sample.** 48,760 pairwise experiments (32,744 decisive); 50 most-evaluated models from 10 developers (Jan–Nov 2025).
- **Context.** Each prompt embedded into a 384-dim vector (`paraphrase-multilingual-MiniLM-L12-v2`), plus topic and token-length covariates.
- **Inference costs.** Per-query energy (kWh) from EcoLogits (Rincé and Banse 2025) estimates—a function of output token length and model complexity.

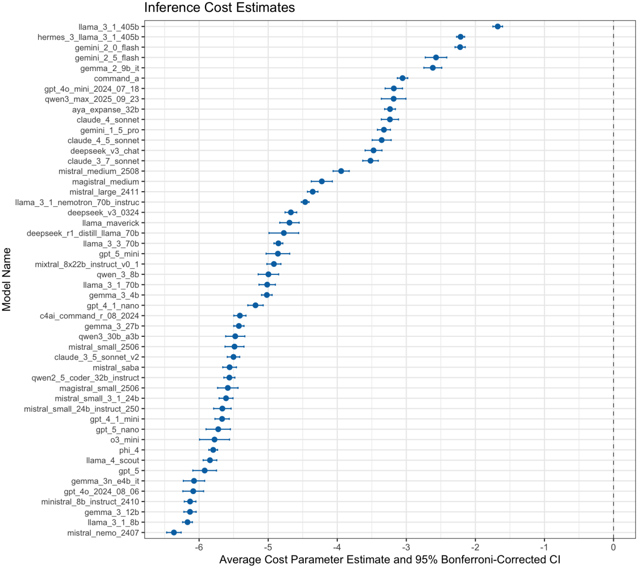
Average Bradley–Terry Estimates: Unconditional vs. Conditional



Distribution of Preference Heterogeneity by Model



Average Inference-Cost Predictions by Model



Results: Cost-Aware Routing Policy

We compare the learned cost-aware policy π^{cost} to the preference-first benchmark $\pi^{\text{pref}}(X) = \arg\max_m \hat{Q}_m(X)$. The target parameters are the policy contrasts in expected log inference cost and expected dominance:

$$\mu_{\text{Cost}} = \mathbb{E}\left[\log c_{\pi^{\text{cost}}}\right] - \mathbb{E}\left[\log c_{\pi^{\text{pref}}}\right], \qquad \mu_{\text{Pref}} = \mathbb{E}\left[Q_{\pi^{\text{cost}}}\right] - \mathbb{E}\left[Q_{\pi^{\text{pref}}}\right].$$

Policy contrast (cost-aware − preference-first)	**Estimate**	**SE**
μ_{Cost} (expected log inference cost)	-0.816^{*}	(0.041)
μ_{Pref} (expected dominance probability)	$+0.001$	(0.002)

- $\mu_{\text{Cost}} = -0.82$ is a $\approx$80% reduction in expected inference energy, while μ_{Pref} is statistically indistinguishable from zero.

$N = 32{,}744$. Judge-clustered standard errors. $^{*}p < 0.05$.

Discussion

Discussion

- **A structural bridge for AI evaluation.** The first application of structural deep learning (Farrell, Liang, and Misra 2021) to preference data takes Bradley–Terry as primitive and recovers context-specific quality with valid orthogonal inference.
- **From rankings to routing.** Estimated dominance probabilities and cost predictions feed an Athey and Wager (2021) policy-learning problem with a tailored orthogonal score, turning leaderboard evaluation into deployable, statistically-backed rules.
- **Gains at negligible downside.** Cost-aware routing cuts expected inference energy by about 80 percent, with a quality change indistinguishable from zero.

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Motivation

- Some empirical settings involve time-varying shocks, where overall exposure to an intervention increases over time
  - e.g., investment/tax incentives, minimum wages, pollution
- It can be challenging to summarize **dynamic** treatment responses due to multiple (continuous) dose exposures
- To inform policy designs, it is often useful to understand causal responses to a single treatment increment in addition to the overall program exposure.
- Conventional strategies in the DiD literature summarize aggregate responses to the entire path of treatment levels or focus on static, one-period-ahead effects
  - de Chaisemartin de Chaisemartin et al. 2022, 2024, 2025

This Paper

1. Develops identification results for dynamic, single-shot treatment effects using difference-in-differences under conditional-on-observables parallel trend assumptions
    - Impulse response to the first non-zero dose over time
2. Constructs two-step estimators that may incorporate machine learning methods for high-dimensional covariate settings
3. Applies the estimators to an empirical application on the labor market effects of the Small Business Administration's 504 loan guarantee program

Framework

Setup

We present results for the 3-period ($T = 3$) case with a single cohort, but generalize to more periods and staggered adoption in the paper.

- Let L_{it} denote unit i's treatment dose in period t
- All units are untreated in the first period ($L_{i1} = 0$)
- A group of treated units receives a first dose in period 2, $L_{i2} \in \mathbb{R}_+$, and a potential second dose in period 3, $L_{i3} \in \mathbb{R}_{\geq 0}$.
- There exists a Never Treated group: $L_{i1} = L_{i2} = L_{i3} = 0$

Potential Outcomes for Cumulative Exposure

- Let $D_{it} = \sum_{l=1}^{t} L_{it}$ denote the cumulative exposure to the treatment intervention as of period t

- Let $\vec{D}_i = (D_{i1}, D_{i2}, D_{i3}) \in \mathbb{R}^3_{\geq 0}$ be the assigned vector of exposures

  - Never Treated units associated with the zero vector, $\vec{0}$
  - Let $G_i = \mathbb{I}(\vec{D}_i \neq \vec{0})$ denote treatment group assignment

- Potential outcomes defined over exposure paths:

$$Y_{it}(\vec{D}_i) = Y_{it}(D_{i1}, D_{i2}, D_{i3})$$

Conventional Estimand: Actual vs Status Quo

1. Actual-vs-Status-Quo ATT, 2-periods from initial treatment:

$$\begin{aligned}\theta^{ATT}(2) &= \mathbb{E}\left[Y_{i3}(0, D_{i2}, D_{i3}) - Y_{i3}(\vec{0}) \mid G_i = 1\right] \\ &= \mathbb{E}\left[Y_{i3} - Y_{i3}(\vec{0}) \mid G_i = 1\right]\end{aligned}$$

   - Aggregates causal responses to all prior changes in treatment
   - Target parameter with conventional strategies (de Chaisemartin and d'Haultfoeuille 2024)

Our Parameter of Interest: Single-Shot

1. Single-Shot ATT, 2-periods from initial treatment:

$$\theta^{SS}(2) = \mathbb{E}\left[Y_{i3}(0, D_{i2}, D_{i2}) - Y_{i3}(\vec{0}) \mid G_i = 1\right]$$

   - Our parameter of interest, but not as easily identified as θ^{ATT}
   - Treated units may receive more than one treatment across the panel, with each dose varying in size, so we don't observe the counterfactual

Identification

Identification Strategy

- We use a **(quasi-)stayer** identification strategy:
  - We assume no treated unit remains at their first exposure, but a non-zero proportion of units remain within a small bandwidth of what would be their single-shot treatment path
- Extension of Stayer/Quasi-Stayer frameworks of:
  - de Chaisemartin & d'Haultfoeuille (2024a): continuous-treatment DiD with stayers
  - de Chaisemartin & d'Haultfoeuille (2024b): intertemporal treatment effects
- All results require an untreated comparison group to proxy for the unexposed counterfactuals of the treated group

Identification Assumptions

Define X_i as a (potentially high-dimensional) vector of time-invariant baseline characteristics for each unit.

Assumption 1 (Overlap). With f the conditional density of period-2 treatments,

$$\mathbb{P}\left(\vec{D}_i = \vec{0} \mid X_i\right) < 1, \quad f\left(D_{i2} = d_2 \mid G_i = 1, X_i\right) > 0 \;\; \forall d_2 \in \mathbb{R}_+$$

Ensures the conditional means used to identify θ^{SS} are well-defined.

Assumption 2 (Conditional Parallel Trends).

$$\forall \vec{d} \neq \vec{0}: \; \mathbb{E}\left[Y_{i3}(\vec{0}) - Y_{i1}(\vec{0}) \mid G_i = 1, X_i, \vec{D}_i = \vec{d}\right] = \mathbb{E}\left[Y_{i3} - Y_{i1} \mid G_i = 0, X_i\right]$$

Never-treated conditional trends serve as the counterfactual for similar-looking treated units.

Identification Assumptions (Continued)

Assumption 3 ("Strong" Conditional Parallel Trends). For any $\vec{d}, \underline{\vec{d}} \in \mathbb{R}^3_{\geq 0}$ with $\vec{d} \neq \underline{\vec{d}}$,

$$\mathbb{E}[Y_{i3}(\vec{d}) - Y_{i3}(\vec{0}) \mid G_i = 1, X_i, \vec{D}_i = \underline{\vec{d}}] = \mathbb{E}[Y_{i3}(\vec{d}) - Y_{i3}(\vec{0}) \mid G_i = 1, X_i, \vec{D}_i = \vec{d}]$$

Conditional on observables, no selection on treatment effects into particular exposure paths.

Assumption 4 (Existence of Conditional Quasi-Stayers).

$$\mathbb{P}\left(|D_{i3} - D_{i2}| < \eta \mid G_i = 1, X_i, D_{i2}\right) > 0, \quad \forall \eta > 0$$

Some treated units remain within a small bandwidth of their single-dose path.

Identification Results

Under Assumptions 1-3 and 4, $\theta^{SS}(2)$ is identified by

$$\theta^{SS}(2) = \mathbb{E}\left[\lim_{\eta \to 0} \mathbb{E}[Y_{i3} - Y_{i1} \mid G_i = 1, X_i, D_{i2}, |D_{i3} - D_{i2}| < \eta] \mid G_i = 1\right] - \mathbb{E}\left[\mathbb{E}[Y_{i3} - Y_{i1} \mid G_i = 0, X_i] \mid G_i = 1\right] \quad (1)$$

Estimation

Parametric Estimation Strategy with Quasi-Stayers

- In this draft, we propose parametric, $\sqrt{n}$-consistent estimators based on conditional mean specifications:

  Assumption 5: Parametric Model

  $$\mathbb{E}[Y_{i3} - Y_{i1} \mid G_i = 1, \vec{D}_i, X_i] = g(\vec{D}_i, X_i; \omega)$$

  - For instance, $g(\vec{D}_i, X_i; \omega) = \alpha + \vec{D}_i^{\mathrm{T}} \beta + X_i^{\mathrm{T}} \lambda$

More Flexible Parameterization

- Varying Coefficient Model:

$$\tilde{g}(\vec{D}_i;\omega(G_i,X_i)) = q_L(\vec{D}_i)^{\mathrm{T}}\beta(G_i,X_i)$$

- Apply ML techniques for estimating these first-stage models (e.g., neural nets, LASSO)
- Use debiasing techniques from the double machine learning literature (Chernozhukov et al. 2021; Chernozhukov et al. 2018)

Automatic Debiasing for ML Estimation

- Let $W_i = (Y_{i3}, Y_{i1}, G_i, \vec{D}_i, X_i)$. If $\tilde{g}$ is linear, we may write $\theta^{SS}(2)$ as a linear functional of first-stage regression functions:

$$\theta^{SS}(2) = \mathbb{E}[\tilde{g}((0, D_{i2}, D_{i2}); \omega(1, X_i)) - \tilde{g}(\vec{0}; \omega(0, X_i)) \mid G_i = 1] = \mathbb{E}[H(W_i, \tilde{g})]$$

- Need to account for first-stage estimation biases from ML — Automatic Debiasing ("AutoDML") (Chernozhukov et al. 2021)
- Estimate α_2^{RR} and $\tilde{g}$, construct the orthogonal influence function:

$$\psi_2 = H(W_i, \tilde{g}) + \alpha_2^{RR}(W_i) \times \left(Y_{i3} - Y_{i1} - \tilde{g}(\vec{D}_i; \omega(G_i, X_i))\right)$$

- No need to derive the influence function, other than knowing $H(W_i, \tilde{g})$. Use a simple loss to obtain α_2^{RR}:

$$\min_{a \in \mathscr{A}} \sum_{i=1}^{n} -2H(W_i, a) + (a(W_i))^2$$

Empirical Application

Empirical Application: SBA 504 Program

- Analyze the regional labor market effects of SBA 504 lending to large-scale local capital investment projects
  - **Previous Work:** county-level, or firm-level analysis (Brown et al. 2017; Craig, Jackson, and Thomson 2009)
  - **Analysis:** Study ZIP code employment trends after post-recession (2010-2019) 504 investment
  - **Challenge:** Multiple within-region 504 investments over time (3 projects of varying size–each \$1.6M, on average)
  - Estimate the single-shot employment impacts, and compare to conventional DiD strategies; specify treated outcome trends as varying coefficients in cumulative dose increments:

$$\mathbb{E}[Y_{i,g+\ell} - Y_{i,g-1} \mid G_i = g, X_i, \vec{D}_i] = \beta_0(X_i) + \sum_{k=0}^{\ell} \beta_k(X_i)(D_{i,g+k} - D_{i,g+k-1})$$

Results

Main Single-Shot Results

Table: Log Employment, 5-Year Estimates

Model	Estimate	Conf. Int.	% Overall
AVSQ	0.0266	[0.0172, 0.0359]	
Single-Shot	0.0114	[0.0035, 0.0193]	43%

Local Multiplier Results (Jobs per Jobs Created)

Normalized single-shot (jobs per 504-backed job):

$$\theta^{SS,\mathrm{Norm}}(g,\ell) = \frac{\theta^{SS}(g,\ell)}{\mathbb{E}[D_{ig} \mid G_i = g]}$$

Normalized AVSQ (de Chaisemartin et al. 2024):

$$\theta^{AVSQ,\mathrm{Norm}}(g,\ell) = \frac{\theta^{AVSQ}(g,\ell)}{\theta^{D}(g,\ell)}, \qquad \theta^{D}(g,\ell) = \mathbb{E}\left[\frac{1}{\ell+1}\sum_{s=g}^{g+\ell} D_{is} \,\middle|\, G_i = g\right]$$

Table: Local Multipliers on Manufacturing Employment, 5-Year Estimates

Estimator	Estimate	95% Conf. Int.
de Chaisemartin & d'Haultfoeuille 2024	0.98	[0.73, 1.23]
Single-shot	2.37	[1.90, 2.83]

Discussion

Discussion

- **Methodology.** The single-shot estimand isolates the dynamic effect of a unit's first dose, which conventional DiD blurs by cumulating the entire exposure path.
- **Identification and Estimation.** A quasi-stayer strategy paired with AutoDML debiasing, gives doubly-robust root-n inference under high-dimensional covariates.
- **Application.** One average 504 project raises local employment about 1.1 percent over five years, roughly 43 percent of the AVSQ effect. Manufacturing employment multipliers understated by standard approaches.

Outline of Talk

1. Chapter 1: Targeted Credit Access and Entrepreneurship
2. Chapter 2: *Ce modèle vous plaît?*
3. Chapter 3: Single-Shot Effects and Semiparametric DiD
4. Conclusion

What's Next?

- Santa Barbara $\rightarrow$ San Diego ✓
- Work as a Senior Data Scientist at Intuit ✓
  - Analyzing experiments (over 50!) relating to product strategy, promotional offers, and pricing
  - Designing experimental plans for our next year offer strategies
  - Extending existing price and targeted discount models using structural models and ML
  - Leveraging AI in almost every task I do as an economist

Thank You

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CDFI Transaction Level Reports (TLR)

- Primary transaction-level source on CDFI lending since 1996
- **What is reported**
  - Every loan or investment linked to a CDFI Fund award over a 3 year reporting window
  - Fields include anonymized awardee ID, year, amount, product type, sector, and borrower address (geocoded to ZIP)

Inferring CDFI Entry from TLR

- **Use of TLR**
  - Treat the first year in which a county records any CDFI TLR lending as the onset of local CDFI access
  - More than one CDFI may be active in a county across the panel
  - Non-awarded CDFIs may be active but not visible in the TLR
- **Interpretation of "entry"**
  - Entry captures the first observable presence of CDFI Fund supported lending in an area
  - Coarse measure of CDFI access and may understate earlier, potentially smaller scale, activity prior to award receipt.

Motivating County-Level Assignment

Average, Square-Approximate County Width: 40 miles